QQ-A-200/14B
27 December 1984
SUPERSEDING
QQ-A-200/14A
June 26, 1974

FEDERAL SPECIFICATION SHEET

ALUMINUM ALLOY BAR, ROD, SHAPES, AND WIRE, EXTRUDED, 7178-T76 (IMPROVED EXFOLIATION RESISTANT)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring the aluminum alloy 7178-T76 bar, rod, shapes, and wire, extruded described herein shall consist of this document and the latest issue of QQ-A-200/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy 7178-T76 bar, rod, shapes, and wire produced by extrusion.

1.2 Classification.

1.2.1 Tempers. The bar, rod, shapes, and wire shall be of the following tempers, as specified (see 6.2): T76, T76510, and T76511. The definition of these tempers shall be in accordance with QQ-A-200/GEN.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-A-200/GEN - Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded; General Specification for

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards

MIL-STD-2154 - Inspection, Ultrasonic, Wrought Metals, Process for

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

G34-72 - Exfoliation Corrosion Susceptibility in 7XXX Series Copper Containing Aluminum Alloys (EXCO Test)

G47 - Determining Susceptibility to Stress-Corrosion Cracking of High-Strength Aluminum Alloy Products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall comform to the requirements specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical composition. 1/

Element	Percent	
	Minimum	Maximum
Zinc	6.3	7.3
Magnesium	2.4	3.1
Copper	1.6	2.4
Chromium	0.18	0.28
Manganese	--	0.30
Iron	--	0.50
Silicon	--	0.40
Titanium	--	0.20
Other elements, each	--	0.05
Other elements, total 2/	--	0.15
Aluminum	Remainder	

1/ Analysis shall routinely be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated or suspected in amounts greater than the specified limits, further analysis shall be made to determine that these elements are not present in excess of specified limits.

2/ The sum of "other" metallic elements, 0.010 percent or more each, expressed to the second decimal place before determining the sum.

3.2 Mechanical properties. Mechanical properties of as-supplied material shall, in the longitudinal (extrusion) direction, conform to the property requirements specified in table II, except as exempted from elongation requirements by QQ-A-200/GEN.

TABLE II. Mechanical properties

Temper	Thickness (bar and shape); diameter (rod and wire); inches	Area, square inches	Tensile strength, minimum, ksi	Yield strength at 0.2 percent offset minimum ksi	Elongation in 2 inches or 4 times D 1/, minimum, percent
T76, T76510, and T76511	Up thru 0.249	Up thru 20	76.0	66.0	7
	0.250 to 0.499	Up thru 20	77.0	67.0	7
	0.500 to 1.000	Up thru 20	77.0	67.0	7

1/ D represents specimen diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

3.3 Internal defects. When specified (see 6.2), rod, bar and shapes shall be ultrasonically inspected (see QQ-A-200/GEN). Acceptance limits shall be as specified in table III .All extrusions in the inspection lot shall be inspected.

TABLE III. Ultrasonic discontinuity acceptance limits. 1/, 2/

Thickness (bar and shapes), inches 3/	Maximum weight of piece, pounds	Maximum width to thickness ratio	Discontinuity class 4/
0.500 - 1.000	600	10 to 1	B

1/ Discontinuities in excess of those listed in table III may be allowed subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in noncritical areas.

2/ Requirements for ultrasonic inspection for sizes, maximum weights and discontinuity classes differing from those shown in table III and for rod shall be specified in the contract provided the techniques and standards are agreed to by the procuring activity and the contractor (see 6.2).

3/ For thickness of a shape, see definition for shape in QQ-A-200/GEN.

4/ The discontinuity class limits are defined in MIL-STD-2154.

3.4 Resistance to exfoliation corrosion and stress-corrosion cracking.

3.4.1 Exfoliation corrosion. Material tested in accordance with 4.2.3 shall not show exfoliation corrosion equivalent to or in excess of that depicted in figure 2 of ASTM G 34-72 for category B. The frequency of tests shall be in accordance with QQ-A-200/GEN.

3.4.2 Stress-corrosion cracking. Extrusions of 0.750-inch thickness or thicker shall exhibit no susceptibility to stress-corrosion cracking under the conditions of the test specified in 4.2.5. The frequency of tests shall be in accordance with QQ-A-200/GEN.

3.4.3 Lot acceptance control criteria. Resistance to exfoliation corrosion and stress-corrosion cracking for each lot of material shall be determined by the following control criteria (see 4.1.2 and 4.2.6).

(a) Determine electrical conductivity and mechanical properties
(b) If the conductivity is 38.0 percent International Annealed Copper Standards (IACS) or higher, and the tensile properties meet the minimum limits specified herein, the material is acceptable

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

(c) If the conductivity is at least 35.0 percent IACS but less than 38.0 percent IACS, and the tensile properties meet the minimum limits specified herein, the material shall either be tested as specified in 4.2.3 or be reprocessed.
(d) If the conductivity is below 35.0 percent IACS, the material is not acceptable and must be reprocessed

3.5 Marking of extrusions. In addition to marking as required in QQ-A-200/GEN, each extrusion shall be marked with its appropriate lot number in at least one location.

4. QUALITY ASSURANCE PROVISIONS (see QQ-A-200/GEN)

4.1 Sampling.

4.1.1 For chemical analysis, mechanical properties, and for exfoliation corrosion and stress-corrosion cracking tests. Samples for chemical analysis, determination of mechanical properties, exfoliation corrosion, and stress corrosion cracking tests shall comply with the applicable requirements of QQ-A-200/GEN.

4.1.2 For conductivity tests. Conductivity shall be measured on the same sample from which specimens are obtained for mechanical property tests.

4.2. Test methods.

4.2.1 Chemical analysis. Samples selected in accordance with 4.1.1 shall be analyzed to determine conformance to 3.1 using the test methods specified in QQ-A-200/GEN.

4.2.2 Ultrasonic inspection. When ultrasonic inspection is specified (see 6.2), extrusion shall be inspected in accordance with MIL-STD-2154 to determine conformance to 3.3.

4.2.3 Exfoliation corrosion test. Samples taken in accordance with 4.1.1 shall be tested for conformance to 3.4.1 using the test method specified by ASTM G 34-72. Full size extrusions or sections of extrusions, sufficiently large to provide visual assessment of attack, shall be used. For all test coupons, 10 percent of the thickness shall be removed by machining one surface. The machined surface shall be the surface evaluated. Chemical milling may be used instead of machining.

4.2.4 Mechanical properties. Samples selected in accordance with 4.1.1 shall be prepared for testing and tested in accordance with QQ-A-200/GEN to determine conformance to 3.2.

4.2.5 Stress-corrosion cracking test. In order to determine conformance to 3.4.2, stress-corrosion cracking tests shall be performed in accordance with ASTM G 47. The specimens shall be stressed in the short transverse direction with respect to grain flow and held at constant strain. The stress level shall be 25 KSi and test duration shall be 20 days.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

4.2.6 Electrical conductivity tests. The sample selected in accordance with 4.1.2 shall be tested for conformance to 3.4.3. Specimen preparation and testing shall be in accordance with QQ-A-200/GEN. The conductivity shall be determined by taking three electrical conductivity readings at random on each sample.

4.3 Test records. The producer shall maintain records of the performance of all lots sampled and tested. Upon request of the procuring activity, such records shall be made available.

5. PREPARATION FOR DELIVERY (see QQ-A-200/GEN)

6. NOTES

6.1 Intended use. This alloy is intended for use where high strength is required, and where exfoliation corrosion resistance superior to that of 7178-T6 is required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Form, Quantity and temper required (see 1.2)
(c) Dimensions required
(d) Requirements for sizes not specifically covered (see QQ-A-200/GEN)
(e) Special end use requirements
(f) Whether ultrasonic inspection of extrusions is required (see 3.3)
(g) Selection of applicable levels of preservation, packaging, and packing (see Section 5 QQ-A-200/GEN)

MILITARY INTEREST:

Custodians

Army-MR
Navy-AS
Air Force-20

Review Activities

Navy-SH
Air Force-99
DLA - IS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
NASA-JFK
DOE-BPA

PREPARING ACTIVITY:

NAVY-AS

DOD Project 9525-0083

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12502 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
QQ-A-200/14B	Aluminum Alloy Bar, Rod, Shapes, and Wire, Extruded, 7178-T76 (Improved Exfoliation Resistant)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:04Z
Check the source to verify that this is the current version before use.